

Curriculum for Undergraduate Degree B.Tech. in Electrical Engineering (w.e.f. AY: 2025-26)

Part I: Introduction, Theme & Category wise Credit Distribution

A. Definition of Credit:

Sl. No.	Description	Credit
1	1 Hr. Lecture (L) Per Week	1
2	1Hr. Tutorial (T) Per Week	1
3	1Hr. Practical/ Lab(P) Per Week	0.5
4	2Hrs. Practical/ Lab(P) Per Week	1

B. Range of Credits:

As per AICTE, a student covering 160 credits during 4 years of studies as per curriculum of the Institute will be eligible to get Under Graduate B.Tech. degree with one major specialization as opted by the student from a list of major specializations as specified in the program curriculum. Over and above, a student will have to earn additional 20 credits (including the credits transferred from SWAYAM platform) and the same shall be mentioned as minor specialization. A student may opt any minor specialization offered by any department of the institute with the constraints that (i) the papers included in that specific specialization should not be same with any of the papers of his/her mandated curriculum and (ii) the student should have the knowledge of the prerequisites w.r.t. the papers of that specialization.

C. Category wise Credit Distribution:

Sl. No.	Category	Credit Allotted	Credit as per AICTE
1	Humanities and Social Sciences including Management Courses	15	12
2	Basic Science Courses	26	32
3	Engineering Science Courses including Workshop, Drawing, Basics of Electrical/ Mechanical/ Computer etc.	21	29
4	Professional Core Courses	55	64
5	Professional Elective Courses relevant to chosen specialization/ branch	15	15
6	Open Elective Courses from other technical and/or emerging subjects	12	9
7	Project Work, Seminar and Internship in Industry or elsewhere	16	15
8	Audit Courses [Environmental Sciences, Induction Training]	0	Non-Credit
Total		160	176

D. Course Code and Definition:

Sl. No.	Course Code	Definitions
1	L	Lecture
2	T	Tutorial
3	P	Practical
4	BS	Basic Science Courses
5	ES	Engineering Science Courses
6	HM	Humanities and Social Sciences including Management Courses
7	PC	Professional Core Courses
8	PE	Professional Elective Courses
9	OE	Open Elective Courses
10	AU	Audit Courses
11	PW	Project/ Internships/ Sessional
12	MN	Minor Courses

E. Courses in different Category:

Humanities and Social Sciences including Management Courses							
Sl. No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	I	HM-HU101	Communication Skill (JeevanKaushal-I)	2	0	0	2
2	I	HM-HU102	Introduction to Indian Knowledge System (Indian Knowledge System-I)	2	0	0	2
3	II	HM-HU203	Inculcation of Human Values and Professional Ethics (MulyaPravah)	2	0	0	2
4	IV	HM-HU401	Leadership and Management Skills (JeevanKaushal-III)	2	0	0	2
5	V	HM-HU502	Aptitude Skill Development-I	1	0	0	1
6	VI	HM-HU601	Aptitude Skill Development-II	1	0	0	1
7	VIII	HM-HU801	Universal Human Values-II (JeevanKaushal-IV)	2	0	0	2
Total Theory				12	0	0	12
Practical/ Sessional/ Audit Course							
1	I	HM-HU191	Communication Skill Laboratory	0	0	2	1

2	V	HM-HU591	Professional Skills (JeevanKaushal-II)	1	0	2	2
<i>Total Practical/ Sessional/ Audit Course</i>				1	0	4	3
Total				13	0	4	15

Basic Science Courses							
Sl. No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	I	BS-M101	Mathematics-I	3	1	0	4
2	I	BS-CH101	Chemistry	3	0	0	3
3	II	BS-M201	Mathematics-II	3	1	0	4
4	II	BS-PH201	Physics	3	1	0	4
5	III	BS-PH301	Electrical Engineering Materials	3	0	0	3
6	IV	BS-BIO401	Biology for Engineers	2	0	0	2
7	IV	BS-M401	Vedic Mathematics (Indian Knowledge System II)	2	0	0	2
8	V	BS-M501	Basic of Indian Astronomy (Indian Knowledge System III)	2	0	0	2
9	VIII	BS-M801	Introduction of Indian Astronomy (Indian Knowledge System IV)	2	0	0	2
Total Theory				23	3	0	26
Practical/ Sessional/ AuditCourse							
1	I	BS-CH191	Chemistry Laboratory	0	0	3	1.5
2	II	BS-PH291	Physics Laboratory	0	0	3	1.5
Total Practical/ Sessional/ Audit Course				0	0	6	3
Total				23	3	6	29

Engineering Science Courses including Workshop, Drawing, Basics of Electrical/ Electronics/ Mechanical/ Computer etc.							
Sl. No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	I	ES-EE101	Basic Electrical and Electronics Engineering	2	1	0	3
2	II	ES-CS201	Programming for Problem Solving	3	0	0	3
3	III	ES-ME301	Engineering Mechanics	3	0	0	3

4	III	ES-EE301	Electrical Engineering Materials	3	0	0	3
<i>Total Theory</i>				11	1	0	12
Practical/ Sessional/ Audit Course							
1	I	ES-EE191	Basic Electrical and Electronics Engineering Laboratory	0	0	4	2
2	I	ES-ME192	Workshop/Manufacturing Practices	0	0	4	2
3	II	ES-CS291	Programming for Problem Solving Laboratory	0	0	4	2
4	II	ES-ME291	Engineering Graphics & Design	0	0	4	2
5	III	ES-CS491	Python Programming	0	0	2	1
<i>Total Practical/ Sessional/ Audit Course</i>				0	0	18	9
Total				11	1	18	21

Professional Core Courses							
Sl. No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	III	PC-EE301	Network Analysis	3	1	0	4
2	III	PC-EE302	Electromagnetic Fields Theory	3	0	0	3
3	III	PC-EE303	Analog Electronics	3	0	0	3
4	III	PC-EE304	Digital Electronics	3	0	0	3
5	IV	PC-EE401	Electric Machine-I	3	1	0	4
6	IV	PC-EE402	Analog and Digital Communication	3	0	0	3
7	IV	PC-EE403	Measurements and Instrumentation	3	0	0	3
8	V	PC-EE501	Electric Machine-II	3	1	0	4
9	V	PC-EE502	Power Transmission and Distribution	3	1	0	4
10	V	PC-EE503	Control System	3	0	0	3
11	VI	PC-EE601	Power Electronics	3	1	0	4
12	VI	PC-EE602	Power System Protection and Analysis	3	1	0	4
13	VII	PC-EE701	Electric Drives	3	0	0	3
Total Theory				39	6	0	45
Practical/ Sessional							
1	III	PC-EE391	Network Analysis Laboratory	0	0	2	1

2	III	PC-EE393	Analog and Digital Electronics Laboratory	0	0	2	1
3	IV	PC-EE491	Electric Machine-I Laboratory	0	0	2	1
4	IV	PC-EC493	Electrical and Electronics Measurement Laboratory	0	0	2	1
5	V	PC-EE591	Electric Machine-II Laboratory	0	0	2	1
6	V	PC-EE592	Power Transmission and Distribution Laboratory	0	0	2	1
7	V	PC-EC593	Control System Laboratory	0	0	2	1
8	VI	PC-EE691	Power Electronics Laboratory	0	0	2	1
9	VI	PC-EE692	Power System Protection and Analysis Laboratory	0	0	2	1
10	VII	PC-EE791	Electric Drives Laboratory	0	0	2	1
<i>Total Practical/ Sessional</i>				0	0	20	10
Total				39	06	20	55

Professional Elective Courses relevant to chosen specialization/ branch							
Sl. No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	VI	PE-EE601	Professional Elective-I	3	0	0	3
2	VI	PE-EE602	Professional Elective-II	3	0	0	3
3	VII	PE-EE701	Professional Elective-III	3	0	0	3
4	VII	PE-EE702	Professional Elective-IV	3	0	0	3
5	VIII	PE-EE801	Professional Elective-V	3	0	0	3
Total Theory				15	0	0	15
Practical/ Sessional							
Total Practical/ Sessional				0	0	0	0
Total				15	0	0	15

Open Elective Courses from other technical and/or emerging subjects							
Sl. No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
1	V	Based on the	Open Elective -I	2	0	0	2

2	VI	papers opted from the list of papers offered by other depts.	Open Elective-II	2	0	0	2
3	VII		Open Elective-III	2	0	0	2
4	VII		Open Elective-IV	2	0	0	2
Total Theory				8	0	0	8
Practical/ Sessional							
1	V	Based on the papers opted from the list of papers offered by other depts.	Open Elective- I Laboratory	0	0	2	1
2	VI		Open Elective-II Laboratory	0	0	2	1
3	VII		Open Elective-III Laboratory	0	0	2	1
4	VII		Open Elective- IV Laboratory	0	0	2	1
Total Practical/ Sessional				0	0	8	4
Total				8	0	8	12

Project Work, Seminar and Internship in Industry or elsewhere							
Sl. No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
Total Theory				0	0	0	0
Practical/ Sessional							
1	II	PW-BS281	Ideation Laboratory	0	0	2	1
2	IV	PW-EE481	Micro Project	0	0	4	2
3	VI	PW-EE681	Mini Project	0	0	4	2
4	VII	PW-EE781	Electrical System Design Laboratory-I	1	0	2	2
5	VII	PW-EE782	Project Stage-I	0	0	6	3
6	VII	PW-EE783	Summer Internship II Evaluation	0	0	2	1
7	VIII	PW-EE881	Electrical System Design Laboratory-II	0	0	2	1
8	VIII	PE-EE882	Project Stage-II	0	0	8	4
9	VIII	PW-EC882	Comprehensive Viva Voce	0	0	0	1
Total Practical/ Sessional				1	0	30	16
Total				1	0	30	16

Audit Courses [Environmental Sciences, Induction Training etc.]							
Sl. No.	Semester	Course Code	Course Name	Contact Hours			Credits
				L	T	P	
Theory							
Total Theory				0	0	0	0
Practical/ Sessional/ Audit							
1	II	AU271	NSS/NCC	2	0	0	0
2	IV	AU-BS471	Environmental Science	2	0	0	0
Total Practical/ Sessional/ Audit Course				4	0	0	0
Total				4	0	0	0